CLASS 10 — SCIENCE FULL SYLLABUS PAPER SET 10

Full Syllabus (Chapters 1–13) | RBSE Board Pattern | Session 2026–27

Time: 3 Hours 15 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 80**

General Instructions:

1. All questions are compulsory.
2. Q. 1–20 are Objective Type (1 mark each).
3. Q. 21–28 are Very Short Answer Type (2 marks each).
4. Q. 29–36 are Short Answer Type (3 marks each).
5. Q. 37–40 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 35 to 40.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 20 × 1 = 20

Choose the correct option and write it in your answer book.

1. The chemical formula of rust is: 1

(a) FeO **(b)** Fe_3O_4 **(c)** $Fe_2O_3 \cdot x\,H_2O$ **(d)** $Fe(OH)_2$

2. The salt NH_4Cl is formed from: 1

(a) KOH and HCl **(b)** NH_4OH and HCl
(c) NH_4OH and H_2SO_4 **(d)** NaOH and HCl

3. Student A says vinegar contains acetic acid; Student B says it contains citric acid. Then: 1

(a) Student B is correct **(b)** Student A is correct
(c) Both are correct **(d)** Both are wrong

4. Which metal melts if kept on the palm? 1

(a) lead **(b)** sodium **(c)** zinc **(d)** gallium

5. Which non-metal is essential for respiration? 1

(a) sulphur **(b)** nitrogen **(c)** oxygen **(d)** chlorine

6. The product formed when ethanoic acid reacts with sodium carbonate includes: 1

(a) H_2 **(b)** CH_4 **(c)** O_2 **(d)** CO_2

7. Which of these is a saturated hydrocarbon? 1

(a) C_3H_8 **(b)** C_2H_4 **(c)** C_3H_6 **(d)** C_2H_2

8. The main nitrogenous waste in humans is: 1

(a) creatine **(b)** uric acid **(c)** urea **(d)** ammonia

9. A student states: “Hormones are secreted by exocrine glands into ducts.” The correct response is: 1

(a) Correct
(b) Only in plants
(c) Incorrect — endocrine glands secrete directly into blood
(d) Cannot be determined

10. Potato is propagated by: 1

(a) spores **(b)** root
(c) leaf **(d)** stem tuber (eyes)

11. Pollen grains are produced in the: 1

(a) anther **(b)** stigma **(c)** ovary **(d)** style

12. The dihybrid cross F2 ratio is: 1

(a) 9 : 3 : 3 : 1 **(b)** 1 : 2 : 1 **(c)** 1 : 1 **(d)** 3 : 1

13. A pure-breeding tall plant has the genotype: 1

(a) tt **(b)** TT **(c)** Tt **(d)** T

14. Magnification $m = -2$ for a mirror means the image is: 1

(a) diminished **(b)** virtual and erect

(c) real, inverted and magnified **(d)** same size

15. A student states: "The colour of the clear sky is blue because blue light has the shortest wavelength of all." The correct response is: 1

(a) Incorrect — violet is shorter; our eyes are more sensitive to blue

(b) Correct

(c) Correct due to oceans

(d) Cannot be determined

16. The resistance of a bulb rated 220 V, 100 W is: 1

(a) 484 Ω **(b)** 0.45 Ω **(c)** 2.2 Ω **(d)** 22 000 Ω

17. The commercial unit of electrical energy is: 1

(a) joule **(b)** kilowatt hour **(c)** volt-ampere **(d)** watt

18. Which of these produces a uniform magnetic field? 1

(a) straight wire **(b)** single loop at its edge

(c) bar magnet near poles **(d)** long current-carrying solenoid

19. An aquarium is an example of a/an: 1

(a) artificial ecosystem **(b)** food chain

(c) biome **(d)** natural ecosystem

20. The increase in the concentration of harmful chemicals at successive trophic levels is called: 1

(a) pollution **(b)** biological magnification

(c) eutrophication **(d)** decomposition

SECTION — B (Very Short Answer Type) 8 × 2 = 16

Answer in about 10–15 words each.

21. Why are decomposition reactions called the opposite of combination reactions? Give an example. 2

22. A student says that diluting an acid increases its H_3O^+ concentration per unit volume. Justify or refute. 2

23. How do amoeba obtain their food? 2

24. What is the function of gibberellins? 2

25. What is pollination? 2

26. Write the genotypic ratio of the F2 generation of a monohybrid cross. 2

27. What is the radius of curvature of a spherical mirror? 2

28. A student says that the magnetic field lines are closed curves. Justify or refute. 2

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SECTION — C (Short Answer Type) — 8 × 3 = 24

Answer in about 30–40 words each.

29. List three observations when dilute hydrochloric acid is added to zinc granules, and write the equation. **3**

30. Give reasons: **3**
(i) Metals are used for making bells.
(ii) Electric wires are made of copper.
(iii) Graphite is used for electrodes.

31. What are the limitations of electrical impulses in nervous coordination? **3**

32. What are STDs? Name two and suggest two preventive measures. **3**

33. A person cannot see objects beyond 1.2 m clearly. Name the defect, the type and power of lens needed. **3**

34. Distinguish between short-circuiting and overloading. **3**

35. What happens to the concentration of H_3O^+ ions when excess base is added to an acid solution? What happens to the pH? Why? **3**

OR

State three uses of common salt (sodium chloride). **3**

36. How do decomposers help in recycling nutrients? Give two examples. **3**

OR

What are the steps you would take to reduce the use of non-biodegradable materials in your daily life? **3**

SECTION — D (Long Answer Type) — 4 × 5 = 20

Answer in about 100–150 words each.

37. (i) Explain image formation by a concave mirror for object at **5**
(a) beyond C
(b) at F
(c) between P and F, with nature of image.
(ii) Which position is used in a shaving mirror?

OR

(i) Describe the image formation by a convex lens for the object at **5**
(a) infinity
(b) beyond 2F
(c) at 2F.
(ii) State the lens formula.

38. **Case study:** Food is cooked in a gas stove. The bottom of a vessel sometimes becomes black. **5**
(i) Why does this happen?
(ii) What does it indicate about the air holes?
(iii) What fuel is used in LPG?
(iv) Why is incomplete combustion dangerous?
(v) Write the equation of complete combustion of butane.

OR

Case study: A student mixed 1 mL ethanol and 1 mL glacial acetic acid with a few drops of conc. H_2SO_4 and warmed it in a water bath. A sweet smell was observed. 5

(i) Name the reaction.

(ii) Write the equation.

(iii) Role of conc. H_2SO_4?

(iv) What happens when the product is treated with NaOH?

(v) Give one use of the product.

39. (i) Derive the expression for equivalent resistance of resistors in parallel. 5

(ii) Why is a parallel arrangement better for household circuits?

OR

(i) Describe an experiment to verify Ohm's law with a circuit diagram. 5

(ii) Draw the expected V–I graph.

(iii) What does its slope represent?

40. (i) Describe the structure and working of the human heart. 5

(ii) Why do fishes have a two-chambered heart?

OR

(i) Describe the human excretory system with a labelled diagram. 5

(ii) Explain the process of urine formation in a nephron.

(iii) How is the amount of urine regulated?

END OF QUESTION PAPER **Total: 80 Marks**

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